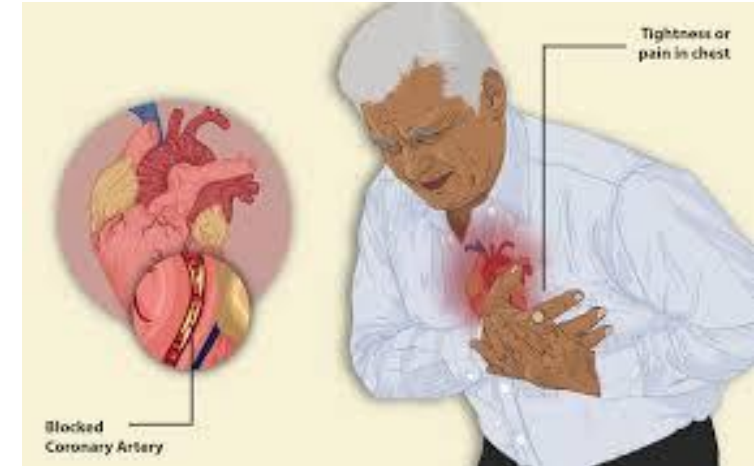


University of Al-Kafeel

College of Dentistry



LEC-8



PHARMACOLOGY OF ANGINA

Asst.Prof.Dr.Yasmeen Ali Hussein

Learning Objectives



By the end of this lecture, you should be able to

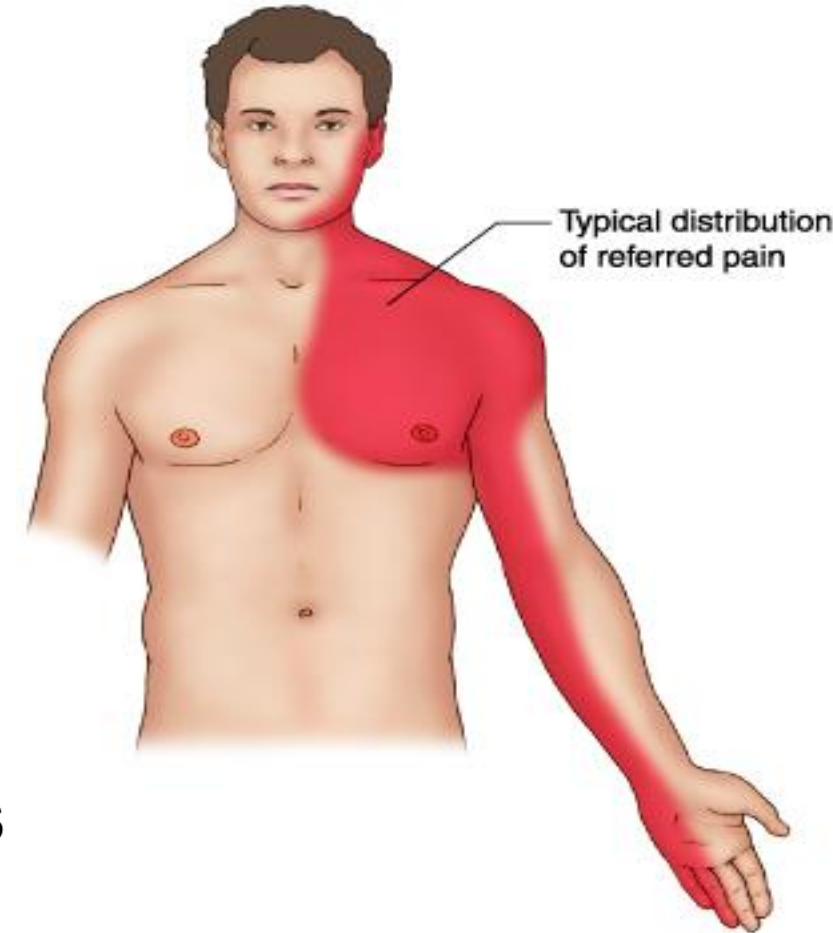
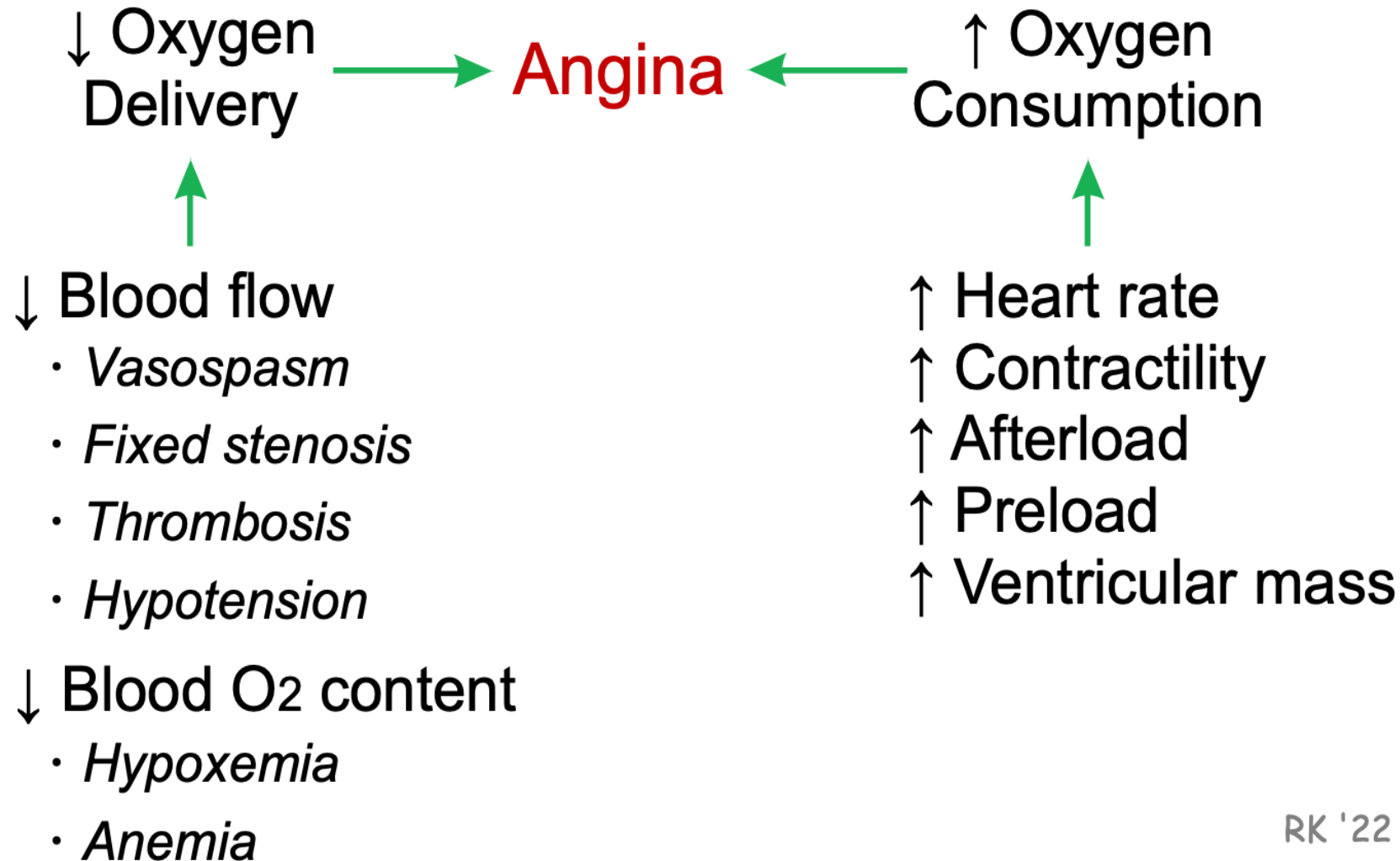
- **Describe** the pathophysiology of angina.
- **Differentiate** between stable angina, unstable angina, variant angina, and microvascular angina.
- **Explain the mechanisms** of action of major drug classes used in angina therapy

Definition of Angina

- It is a clinical syndrome resulting from imbalance between the myocardial oxygen requirement and myocardial oxygen supply. The cause is myocardial ischemia and the most common cause is coronary atherosclerosis.



Pathophysiology of Angina



RK '22



RISK FACTORS FOR ANGINA

1. NON-MODIFIABLE RISK FACTORS



AGE

Progressive arterial changes.



FAMILY HISTORY

Genetics & early heart attacks)



GENETIC / ETHNIC PREDISPOSITION

Higher natural risk



UNHEALTHY DIET

High fat, salt, sugar

2. MODIFIABLE LIFESTYLE RELATED RISK FACTORS



PHYSICAL INACTIVITY

Low fitness, ↑BP, obesity



SMOKING

Vessel damage, plaque



OBESITY / EXCESS WEIGHT

↑ Heart workload, cholesterol, BP



ALCOHOL MISUSE

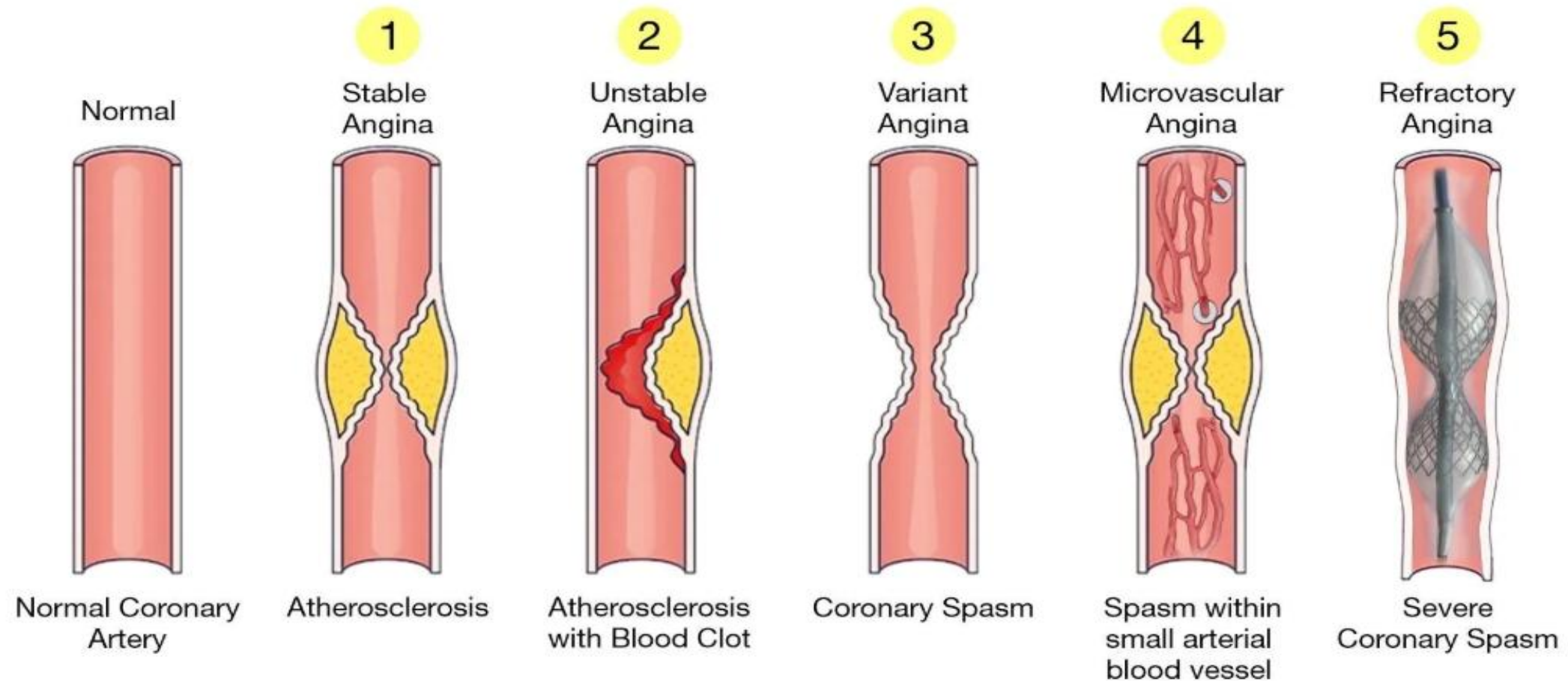
↑ BP, weakens heart,

These CANNOT be changed.

These CAN be changed



Types of Angina



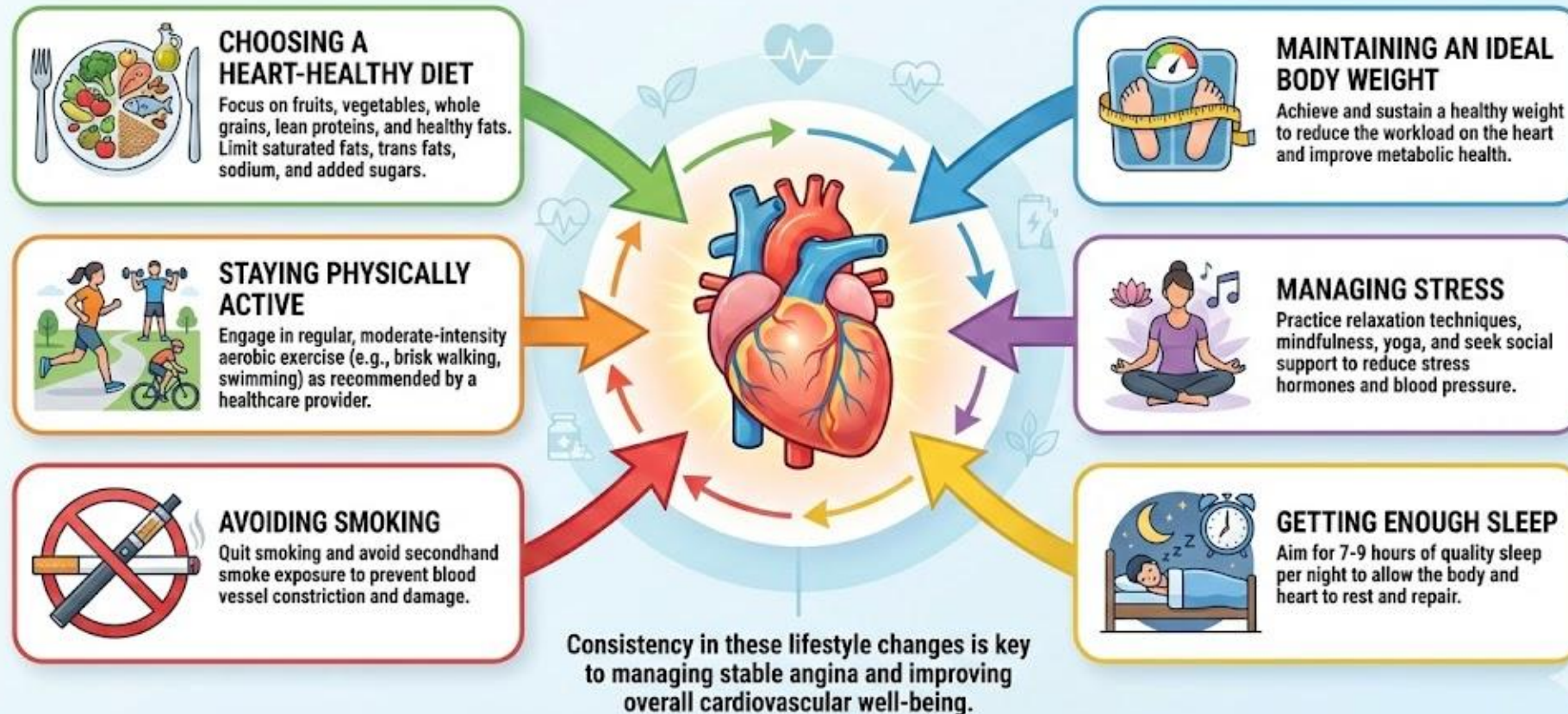


TREATMENT OF ANGINA



NONPHARMACOLOGICAL MANAGEMENT OF ANGINA

Lifestyle modifications for stable symptoms to improve heart health and reduce angina frequency.



PHARMACOLOGIC MANAGEMENT OF ANGINA

Classically used drugs for angina pectoris



NITRATES

Reduce preload and myocardial oxygen demand



BETA-BLOCKERS

Decrease heart rate and contractility



CALCIUM CHANNEL BLOCKERS

Improve coronary blood flow



ANTIPLATELET AGENTS

Reduce platelet aggregation

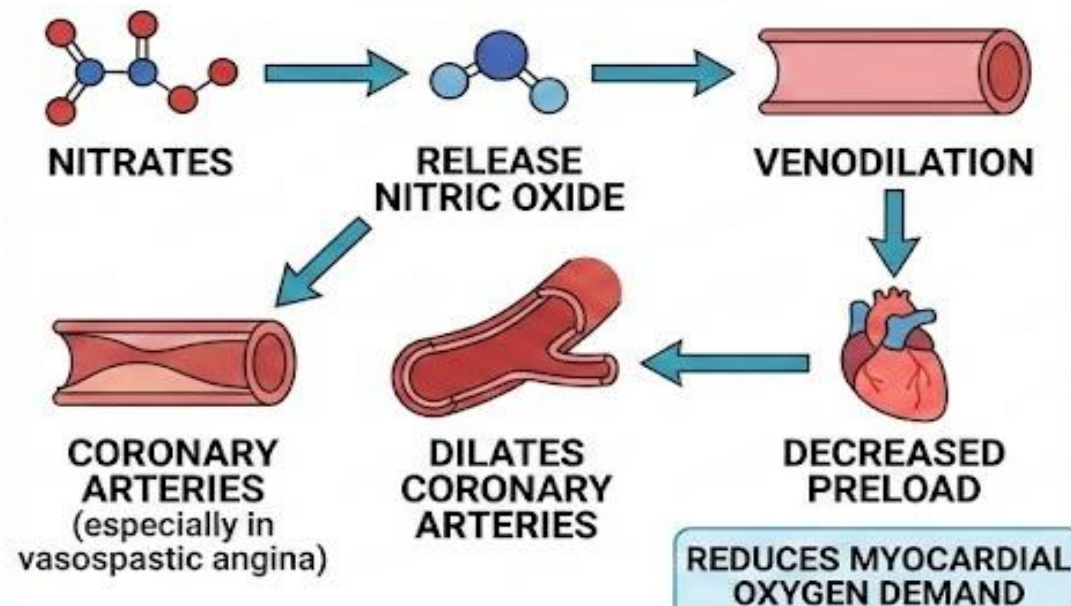


NITRATES: Mechanism, Uses, and Safety

USES

- Acute relief of angina attacks
- Prophylaxis before exertion
- Variant (Prinzmetal) angina

MECHANISM



EXAMPLES

NITROGLYCERIN

ISOSORBIDE
MONONITRATE

ISOSORBIDE
DINITRATE

KEY ADVERSE EFFECTS

- Headache
- Reflex tachycardia
- Hypotension
- Tolerance with continuous use

CONTRAINDICATIONS

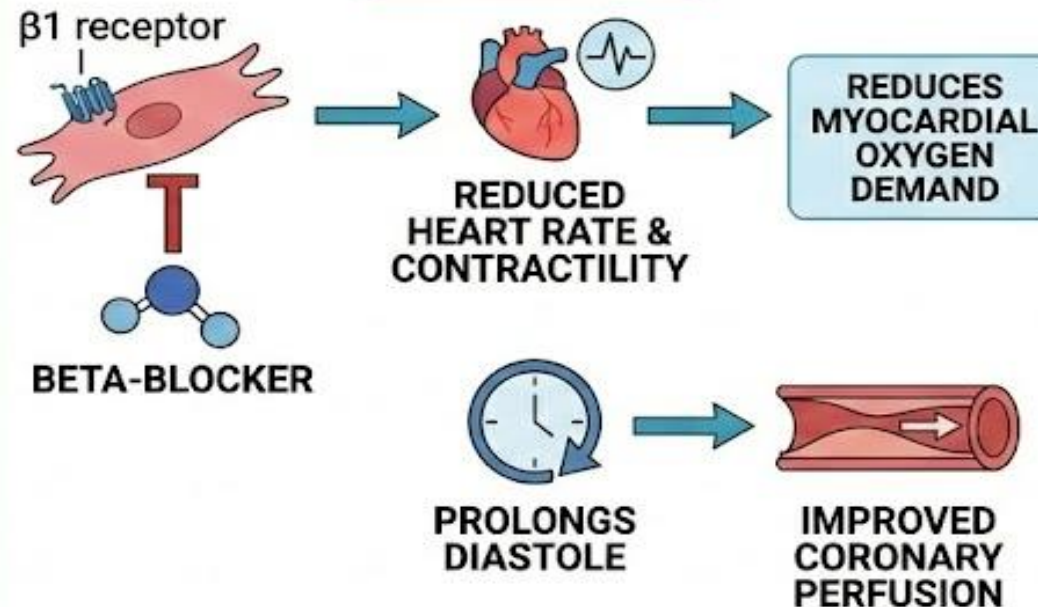
- Concomitant use with PDE-5 inhibitors (e.g., sildenafil)
- Severe hypotension

BETA-BLOCKERS: Mechanism, Uses, and Safety

USES

- Stable angina
- Post-MI secondary prevention
- Hypertensive patients with CAD

MECHANISM



KEY ADVERSE EFFECTS

- Bradycardia
- Fatigue
- Bronchospasm (non-selective agents)

CONTRAINDICATIONS

- Severe asthma
- Bradycardia, AV block
- Acute decompensated heart failure

EXAMPLES

METOPROLOL

ATENOLOL

BISOPROLOL

CARVEDILOL

CALCIUM CHANNEL BLOCKERS (CCBs): Mechanism, Uses, and Safety

USES



- Stable angina (alternative or add-on to β -blockers)



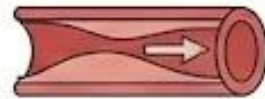
- Variant angina (first-line)



- Hypertensive CAD patients

MECHANISM

DIHYDROPYRIDINES
(e.g., amlodipine)



VASODILATION

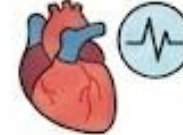


↓ AFTERLOAD

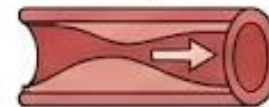


↑ CORONARY BLOOD FLOW

NON-DIHYDROPYRIDINES
(verapamil, diltiazem)



↓ HEART RATE AND CONTRACTILITY



VASODILATION

KEY ADVERSE EFFECTS



- Peripheral edema (DHPs)



- Constipation (verapamil)



- Bradycardia, AV block (non-DHPs)

CONTRAINDICATIONS



- HFrEF (verapamil/diltiazem)



- Severe bradycardia

EXAMPLES

AMLODIPINE

DILTIAZEM

VERAPAMIL

Antiplatelet Drugs

USES



- All patients with stable angina unless contraindicated

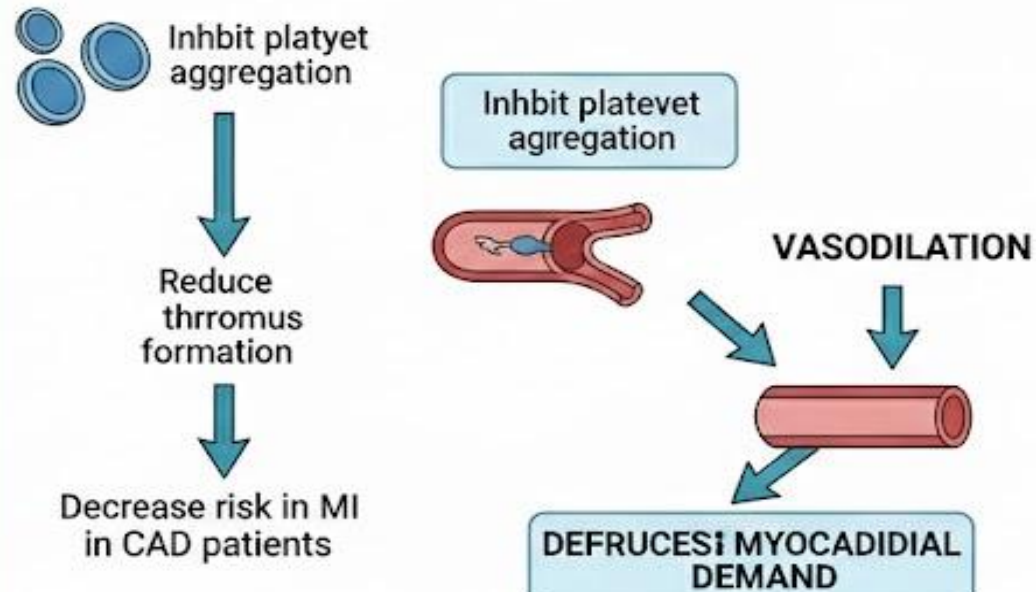


- Secondary prevention after MI or PCI



- Hypertensive CAD patients

MECHANISM



KEY ADVERSE EFFECTS



- GI bleeding



- Bruising



- Hypotension



- Dyspnea (ticagrelor)

CONTRAINDICATIONS



- Active bleeding



- Aspirin hypertension

EXAMPLES

ASPIRIN
(first-line)

CLOPIDOGREL
(alternative or dual therapy)

TICAGRELOR
(mainly ACS)

References



- **Lippincott's** Illustrated Reviews: Pharmacology, 8th Edition. 2022
- **Katzung's** Basic & Clinical Pharmacology, 14th Edition. 2018
- **Rang and Dale's** Pharmacology, 8th Edition. 2021



THANK
YOU!

